# Xiyang Wu

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## Research Profile

Ph.D. candidate specializing in reinforcement learning and post-training for multimodal foundation models and self-improving agents. Extensive hands-on experience designing and running end-to-end SFT/RFT experiments, from multimodal data curation, synthetic-data generation, and rollout pipelines to curriculum and reward design and distributed multi-GPU training. Leads research on long-horizon agent memory and continual skill learning, physics-aware video reasoning and world-model evaluation, and reliability and red-teaming of VLM/VLA systems. First-author publications at NeurIPS, CVPR, EMNLP, IROS, and CoRL.

## Education

**University of Maryland** — College Park, MD
*Ph.D. in Electrical and Computer Engineering, GAMMA Laboratory* — *Aug 2021 – Dec 2026 (Expected)*
Advisor: Dinesh Manocha GPA: 3.83/4.00

**Georgia Institute of Technology** — Atlanta, GA
*M.S. in Electrical and Computer Engineering, CORE Robotics Laboratory* — *Aug 2019 – May 2021*
Advisor: Matthew Gombolay GPA: 4.00/4.00

**Tianjin University** — Tianjin, China
*B.Eng. in Measuring and Controlling Technologies and Instruments (Honors Class)* — *Sep 2015 – Jul 2019*
Advisor: Xiaodong Zhang GPA: 3.85/4.00

## Selected Publications

(*) indicates equal contribution.

1. **Xiyang Wu**, Zhichao Liu, Shengxin Zhang, Zongxia Li, Ming Li, Tianyi Zhou, Dinesh Manocha. Failure Recovery via Belief Updates and World Modeling for Long-Horizon Agents. *arXiv:2610.18082, 2026*, Link.
2. **Xiyang Wu**, Zongxia Li, Shengxin Zhang, Zhichao Liu, Dinesh Manocha. ASSEMBLE: Atomic Skills for Evidence-Grounded Video Reasoning. *arXiv:2609.32466, 2026*, Link.
3. **Xiyang Wu**, Zongxia Li, Guangyao Shi, Alexander Duffy, Tyler Marques, Matthew Lyle Olson, Tianyi Zhou, Dinesh Manocha. Co-Evolving LLM Decision and Skill Bank Agents for Long-Horizon Tasks. *arXiv:2604.20987, 2026*, Link, Code, Project Page, Model.
4. **Xiyang Wu**, Guangyao Shi, Qingzi Wang, Zongxia Li, Amrit Singh Bedi, Dinesh Manocha. SABER: A Stealthy Agentic Black-Box Attack Framework for Vision-Language-Action Models. *arXiv:2603.24935, 2026, IROS 2026*, Link.
5. **Xiyang Wu**, Zongxia Li, Jihui Jin, Guangyao Shi, Nilotpal Sinha, Gouthaman KV, Vishnu Raj, Fan Du, Dinesh Manocha. MASS: Motion-Aware Spatiotemporal Grounding for Physics Reasoning and Comprehension in Vision-Language Models. *arXiv:2511.18373, CVPR 2026*. Link.
6. **Xiyang Wu***, Zongxia Li*, Yubin Qin, Guangyao Shi, Hongyang Du, Dinesh Manocha, Tianyi Zhou, Jordan Lee Boyd-Graber. VideoHallu: Enhancing Multimodal Synthetic Video Understanding via Reinforcement Curriculum Learning. *arXiv:2505.01481, NeurIPS 2025*. Link, Code, Project Page, Dataset.
7. **Xiyang Wu**, Souradip Chakraborty, Ruiqi Xian, Jing Liang, Tianrui Guan, Fuxiao Liu, Brian Sadler, Dinesh Manocha, Amrit Singh Bedi. Highlighting the Safety Concerns of Deploying LLMs/VLMs in Robotics. *arXiv:2402.10340, IROS 2025*. Link, Code, Project Page.
8. **Xiyang Wu**, Tianrui Guan, Dianqi Li, Shuaiyi Huang, Xiaoyu Liu, Xijun Wang, Ruiqi Xian, Abhinav Shrivastava, Furong Huang, Jordan Lee Boyd-Graber, Tianyi Zhou, Dinesh Manocha. AUTOHALLUSION: Automatic Generation of Hallucination Benchmarks for Vision-Language Models. *arXiv:2406.10900, EMNLP 2024*. Link, Project Page.
9. **Xiyang Wu***, Chak Lam Shek*, Wesley A. Suttle, Carl Busart, Erin Zaroukian, Dinesh Manocha, Pratap Tokekar, Amrit Singh Bedi. LANCAR: Leveraging Language for Context-Aware Robot Locomotion in Unstructured Environments. *IROS 2024*. Link, Project Page.
10. **Xiyang Wu**, Rohan Chandra, Tianrui Guan, Amrit Singh Bedi, Dinesh Manocha. Intent-Aware Planning in Heterogeneous Traffic via Distributed Multi-Agent Reinforcement Learning. *CoRL 2023 (Oral)*, *IROS 2023 MRS Workshop (Best Paper Award)*. Link, Code.

Complete publications: Google Scholar Project details: wuxiyang1996.github.io.

## Industry Research Experience

**NEC Laboratories America, Inc.** Princeton, NJ
*Research Intern* *Mentors: Deep Patel, Iain Melvin, Martin Renqiang Min* *Jun 2026 – Aug 2026*

- **Streaming Video-LLMs with Long-Term Memory.** Own the end-to-end architecture for long-horizon streaming video understanding, designing online multimodal memory construction, indexing and retrieval, and agentic multi-hop temporal reasoning over day-scale egocentric video under bounded context.

**Good Start Labs** Remote
*Research Collaboration* *Jan 2026 – Present*

- **Self-Improving Agents and Continual Skill Learning.** Led COS-PLAY; developed a co-evolving decision and skill-bank agent architecture and designed trajectory-to-skill mining, verification, and retrieval/composition, enabling an 8B model to outperform strong frontier-LLM baselines across multiple long-horizon environments.

**Dolby Laboratories, Inc.** Atlanta, GA
*Research Intern* *Mentors: Jihui Jin, Gouthaman KV, Vishnu Raj, Nilotpal Sinha* *May 2025 – Aug 2025*

- **Physics-Aware Synthetic Video Understanding.** Led the development of MASS, a model-agnostic spatiotemporal grounding and post-training framework for Video-LLMs; architected a multimodal pipeline that fuses depth, object grounding, and motion tracks, then applied SFT and curriculum-based reinforcement fine-tuning to improve physics reasoning on real and synthetic video (CVPR 2026).

## Research Experience

**GAMMA Laboratory, University of Maryland** College Park, MD
*Research Assistant* *Advisor: Dinesh Manocha* *Sep 2022 – Present*

- **Self-Improving Agents and Reinforcement Learning.** Lead research on continual agent improvement through reusable skill acquisition, reinforcement learning, and multimodal feedback for long-horizon decision-making.
- **Multimodal Reinforcement Learning and Post-Training.** Co-led VideoHallu from problem formulation through post-training; established synthetic video as a diagnostic and training signal, defined the capability taxonomy, and designed the data curation and curriculum reinforcement fine-tuning (RFT) pipelines to improve synthetic-video reasoning without degrading real-video performance (NeurIPS 2025); led MASS, integrating depth, visual grounding, and motion cues with SFT and curriculum RFT for physics-aware video reasoning (CVPR 2026).
- **Reliability, Safety, and Red-Teaming of Multimodal Models.** Conceived and led SABER end to end, formulating agentic black-box attacks on VLA models (IROS 2026); led AUTOHALLUSION, automating verifiable counterexample generation to expose VLM hallucinations (EMNLP 2024); led an adversarial evaluation of LLM/VLM failure modes in robotics (IROS 2025).
- **Reinforcement Learning for Embodied Intelligence.** Led iPLAN, formulating distributed multi-agent reinforcement learning (MARL) that infers nearby agents' intents from local observations and decouples behavioral decision incentives from motion-planning rewards (CoRL 2023 Oral; IROS MRS Best Paper Award); co-led LANCAR's language-based context interface for robot locomotion under ambiguous human descriptions (IROS 2024).

## Skills

- **Programming & ML:** Python, PyTorch, C/C++, ROS, MATLAB, shell scripting
- **Foundation Model Training:** Distributed multi-GPU training, rollout generation, instruction tuning (SFT), DPO, GRPO, RLVR, curriculum learning, reward design, vLLM, verl, TRL
- **Agentic AI:** Self-improving agents, long-horizon planning, multimodal memory and retrieval, skill discovery and composition, continual skill learning, multi-agent reinforcement learning
- **Multimodal & Embodied AI:** Vision-language models (VLMs), vision–language–action models (VLAs), video understanding, visual grounding, embodied planning, motion and behavior planning, physics reasoning
- **Data, Evaluation & Safety:** Multimodal data pipelines, data curation and filtering, synthetic data generation, automated evaluation, hallucination analysis, adversarial red-teaming

## Awards & Academic Service

- **Best Paper Award:** IROS 2023 Multi-Robot Systems Workshop
- **Journal Reviewer:** *IEEE Access*, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *Journal of Medical Internet Research (JMIR)*, *IEEE Robotics and Automation Letters (RA-L)*
- **Conference Reviewer:** *NeurIPS*, *CVPR*, *CoRL*, *ICRA*, *IROS*, *EMNLP*, *NAACL*, *ACL*
- **Program Committee Member:** CoCoMARL Workshop at RLC